

Name:

Date:

Period:

Simple Harmonic #1

1. You hang 0.45 kg off of the end of a 1.5-m long string. Calculate the period of the pendulum.
2. A pendulum swings back and forth 20 times in 30 seconds. What is the length of the pendulum?
3. A 0.7-meter long pendulum is taken to the moon, where it has a period of 4.2 seconds. What is the acceleration of gravity on the moon?

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4. What is the period of a pendulum that has a mass of 0.25 kg and a length of 2 meters?

5. What is the length of a pendulum that has a period of 5.2 seconds?

6. What are the two things that affect the period of a pendulum?

7. Why does amplitude *not* affect the period of a pendulum?

8. Why does mass *not* affect the period of a pendulum?

9. A flautist in the marching band makes 120 steps in one minute (60 seconds). What is the flautist's period of steps?